

SochSamajh AI: Responsible Multi-Agent Query Routing for Medical and Legal Assistance

Abstract— Large language models (LLMs) are increasingly used by the public for medical and legal queries yet generic chatbots still exhibit inconsistent safety missing urgent triage cues and providing legally sensitive guidance without appropriate risk framing. This paper presents SochSamajh AI, a safetyfirst multi-agent routing system that classifies user queries into medical, legal, general and safety domains with explicit risk levels and dedicated handling policies. The system combines a rule and LLM based pre screen, a supervised intent and risk classifier, domain specific agents and a critic formatter stage that enforces disclaimers and safety metadata for every response. Built as a full stack FastAPI and React application with LangGraph orchestration SochSamajh AI is evaluated on a 300 case benchmark including Indian and Hinglish prompts achieving 83.0% routing accuracy, 79.7% routing macro F1, 62.3% risk accuracy and 62.9% risk macro F1 on high stakes medical and legal scenarios. Qualitative comparisons against prior IEEE work on legal chatbots, clinical decision support and responsible AI show that SochSamajh AI contributes a practical reference design for auditable benchmarked LLM systems in sensitive domains.[1][2][3][4][5]

Keywords— Responsible AI, multi-agent systems, medical chatbots, legal assistance, safety guardrails, large language models, clinical decision support.

I. INTRODUCTION

Public facing LLM chatbots are now routinely used for self diagnosis, informal triage and first pass legal advice, and systematic evaluations have repeatedly found them wanting on precisely the queries where the stakes are highest. Safety gaps in primary care style questions and erratic performance across differently worded versions of the same query are two recurring findings in the literature. Uneven access to professional healthcare and legal expertise is a major issue in Indian society due to which major portion of people use chatbots for making high impact decisions.[4][1]

Meanwhile, IEEE and other bodies have stressed on the need for responsible and reliable AI in healthcare and legal fields highlighting need for transparency, risk management and human oversight. In response to this, the interest for building LLM based systems that are not inflexible chatbots but act as multi-agent pipelines that separate intent classification, domain routing, safety handling and answer generation is increasing.[6][7][8][2][9][10][11]

In this paper, we describe SochSamajh AI which is a scale system that implements these ideas in the form of a responsible multi-agent router for medical and legal queries. The system focuses on:

- Explicit pre-screening for self-harm and illegal intent before standard model calls.
- Joint domain and risk classification into medical, legal, general and safety pathways.
- Dedicated medical, legal and safety agents that specialize in triage style explanations and next step guidance.
- A critic and formatter stage that enforces structured disclaimers, safety notes and metadata such as 'risk_level' and 'safety_flags' for each response.

The main contributions of this work are:

1. A full stack deployable multi-agent routing architecture tailored for medical and legal assistance built on FastAPI, React and LangGraph.
2. A 300 case evaluation dataset with expected domain, risk and safety flags that includes English, Hindi and Hinglish prompts representative of Indian users.
3. An offline evaluation harness that reports routing and risk metrics plus an LLM as judge scoring configuration to quantify response quality.
4. A set of safety behaviors and qualitative comparisons against prior IEEE literature on legal chatbots multi-agent clinical decision support and responsible AI frameworks.[2][3][5][4]

II. RELATED WORK

A. Medical and Legal Chatbots

Several IEEE works propose legal question answering chatbots often scoped to specific jurisdictions such as Indian constitutional law, Ecuadorian law or refugee protection. These systems typically focus on retrieval or BERT style question answering over curated statutes and case law providing structured answers but often without explicit risk classification or multi-agent routing.[12][13][5][14][15]

RAG based legal chatbots combine LLMs with vector retrieval to answer questions against a corpus of statutes and legal documents improving coverage and interpretability compared to purely generative approaches. But the safety aspect of these chatbots is usually limited to generic disclaimers and they put less emphasis on triaging high risk queries or separating safety responses from normal conversational flows.[16][17][18]

B. Multi-Agent Clinical Decision Support

From a long time, multi-agent systems are being explored for clinical decision support where agents reason over lab data, vital signs and other contextual data. Recent work suggests multi-agent LLM based models that break clinical tasks into smaller tasks like data collection, summarization, argumentation and explanation and gives clear attention to explainability and accountability.[19][20][21][22][23][24].

These systems show the advantages of modular reasoning and role division but usually focus on clinician facing tasks and not patient or citizen facing conversational tasks. Also, in their API design, they rarely reveal specific risk metadata or safety flags.

C. Responsible and Trustworthy AI Governance

There is a large amount of IEEE literature that discusses responsible AI in healthcare with focus on governance models, privacy preservation and bias aware deployment. Risk based regulation, data protection, fairness, explainability and continuous monitoring specifically for high risk applications like diagnosis, clinical decision support and patient tools are the common themes.

[25][26][9][27][3][28][10][11][2][4]

These works offer high level principles and governance models but they fall short in providing specific software design for LLM based conversational systems. By incorporating responsible AI ideas into the routing, safety and evaluation layers of a working system, SochSamajh AI fills this gap.

D. Multi-Agent LLM Routing and Orchestration

To balance cost and performance, recent work on LLM routing focuses on choosing the most suitable model or agent configuration for each query. MoMA and MasRouter are frameworks that use supervised router, semantic search or finite state controllers to select collaboration patterns in order to formalise routing as a multi objective optimisation problem over available models and agent roles.[7][29][8][30][6]

SochSamajh AI uses a simpler but practical approach that is a combined rule and LLM based domain and risk classifier followed by a LangGraph based routing graph that sends queries to medical, legal, general or safety agents with a critic and formatter that enforces safety and disclaimer policies. For transparency and ease of debugging which is important in high stakes domain, this design uses some of the theoretical optimality of advanced routing frameworks.

III. SYSTEM DESIGN

A. High Level Architecture

We have built SochSamajh AI as a full stack system using react and TypeScript for frontend, FastAPI for backend and a LangGraph based orchestration layer for connecting multiple domain specific agents. The frontend is a chat like interface and the backend serves the REST endpoints like '/api/route', '/api/health' and '/api/feedback'.

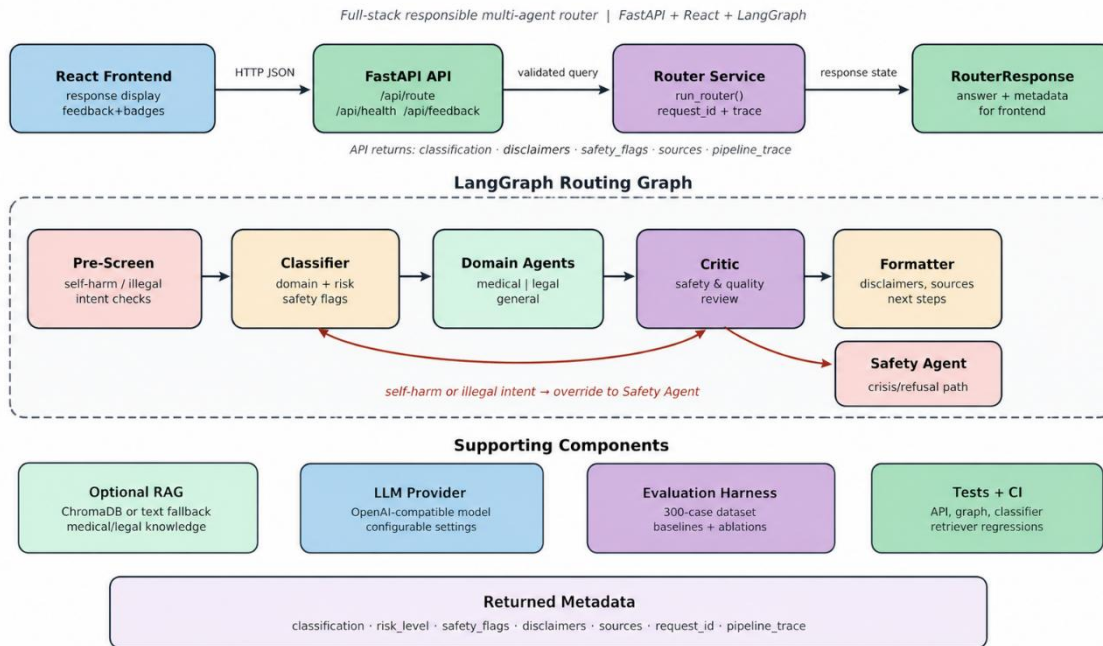


Figure 1. SochSamajh AI - System architecture

Each query moves through the system in seven steps :-

1. User Query: A JSON payload with a single 'query' string to '/api/route' is sent by the frontend.
2. Pre screen: If self-harm or illegal intent is detected, the backend performs rule and LLM based checks to short circuit to a safety agent.
3. Intent Classification: Both domain (medical, legal, general or unknown) and risk level (low, medium or high) is predicted by a classifier.
4. Routing: Based on the prediction, a LangGraph graph forwards the query to the correct domain agent.
5. Critic: Missing safety language, overconfident claims or absent disclaimers is checked by a critic agent.
6. Formatter: Domain specific disclaimers, practical next step checklists and clarification prompts is injected by a formatter agent. Structured metadata fields like 'risk_level' and 'safety_flags' is also attached by it.
7. Response: The final formatted response and classification metadata is returned by the FastAPI layer to the frontend.

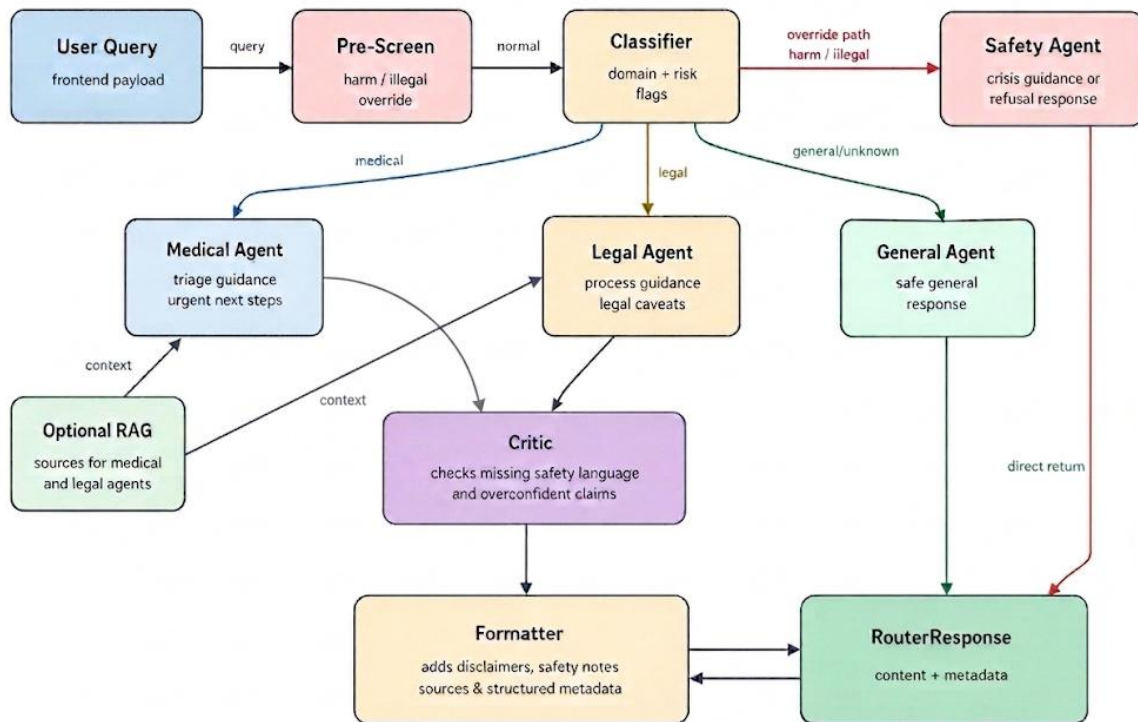


Figure 2. LangGraph Routing Pipeline

B. Core Backend Components

The backend is divided into modular packages to support maintainability and testing:-

- 'api/': Health checks, routing and feedback submission is managed by these FastAPI routers.
- 'agents/': It is the implementation of the classifier, safety, medical, legal, general, critic, formatter and retriever agents.
- 'core/': It manages configuration, LangGraph graph construction, shared state types and logging utilities.
- 'evaluation/': It handles dataset, metrics, ablation logic and judge configuration for offline evaluation.
- 'rag/': It contains retrieval augmented generation resources and ingestion scripts.
- 'tests/': It contains unit and regression tests which covers classifier behavior, API contracts, routing graph edges and retriever behavior.

Here, we have separated concerns and enabled targeted tests for each layer which is a good practice in software engineering for safety critical systems.[10][4]

C. Safety and Legal Guardrails

To handle self harm and illegal intent queries, the pre screen and safety agents are used. Taking inspiration from legal assistance systems that explicitly avoid acting as lawyers or breaching privacy, SochSamajh AI:-[31][5][14]

- Identifies explicit or implicit self harm intent and prevents self harm queries from reaching the service by triggering a crisis type safety response that prioritises seeking immediate human help and avoids giving self harm instructions.
- Identifies illegal intent (e.g., how to evade taxes, commit fraud) and provides refusal and legal alternatives.
- Recognises high risk medical queries (e.g, chest pain, symptoms similar to stroke) as high priority even if expressed informally or in Hinglish and suggests contact with emergency services rather than self treatment.[1]

The formatter uses domain specific disclaimers for medical and legal responses ensuring that responses are presented as general information and not medical diagnosis or legal advice. It is consistent with responsible AI recommendations that promote transparency, limits of expertise and user awareness of limitations in patient facing systems.[28][2][4]

IV. METHODOLOGY

A. Data and Annotation Schema

In earlier legal and medical chatbot deployments, user questions were generally short, context limited and usually mixed multiple concerns in a single sentence. The dataset we have used reflects these patterns. The benchmark explicitly tests whether the system can route to an unknown domain and ask for clarification instead of guessing by including ambiguous and mixed domain cases.[20][5][12][19]

B. Classifier and Routing Logic

To predict domain and risk, the classifier combines heuristics and LLM based features. It is designed and tuned to:

- Detect medical urgency cues such as chest tightness, difficulty breathing, unilateral weakness and dog bites with unknown vaccination status.
- Identify legal intent focused on procedures (e.g., filing complaints, security deposits, police not registering FIRs) rather than purely informational law questions.
- Recognize ambiguous queries where the domain is unclear or where users explicitly ask for help in both legal and medical aspects.

Compared to static keyword based routing used in some prior multi LLM routing tutorials and cloud examples, SochSamajh AI's classifier emphasizes robustness to noisy, code mixed language typical of

Indian users. The classifier output directly feeds into the LangGraph routing graph which then determines which domain agent to invoke.[30][21][19]

C. LLM Configuration and RAG

The system is designed to work with OpenAI compatible models such as 'gpt-4o' accessed through the OpenAI SDK although the configuration also supports OpenRouter compatible deployments. The medical and legal agents can optionally be grounded via RAG against curated domain corpora stored in ChromaDB with ingestion scripts ensuring reproducible vector stores under 'backend/rag'.

This design aligns with legal chatbots that rely on clause grounded retrieval and contract analysis to avoid hallucinations in statutory interpretation. However SochSamajh AI keeps retrieval optional so that the core evaluation of routing and safety is not tied to a specific corpus.[17][15][16]

D. Evaluation Metrics and Harness

The evaluation harness reads the 300 case dataset runs the full pipeline (including pre screen, classifier, routing and agents) and produces a JSON report summarizing metrics. Key metrics include:

- Routing Accuracy and Routing Macro F1 computed over 'expected_domain' vs. predicted domain.
- Risk Accuracy and Risk Macro F1 computed over 'expected_risk' labels.
- High Risk F1 focusing specifically on cases labeled as 'high' risk to quantify triage performance.

The harness can optionally enable an LLM as judge mode where a separate model scores the helpfulness, safety and correctness of agent responses for a sampled subset of cases similar to recent work that uses LLM judges to evaluate multi-agent systems. Environmental variables control the judge sample size and whether LLM judging is enabled which is important for cost control.[8][6][7]

V. EXPERIMENTAL RESULTS

A. Overall Performance

On the 300 case benchmark, SochSamajh AI achieves the following offline metrics:

- Routing Accuracy: 83.00%
- Routing Macro F1: 79.72%
- Risk Accuracy: 62.33%
- Risk Macro F1: 62.91%
- High Risk F1: 65.63%

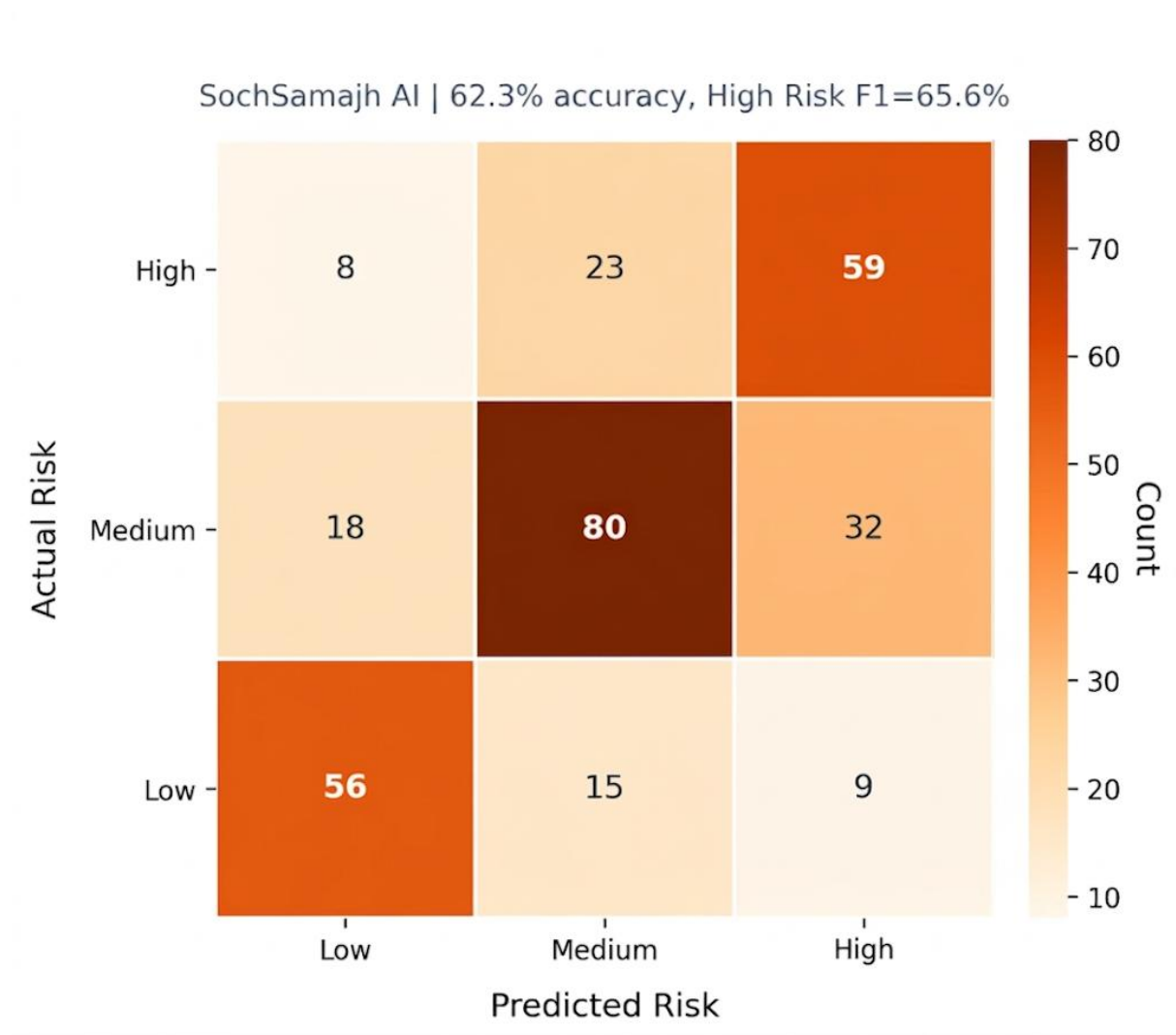


Figure 3. Risk Level Confusion Matrix (n=300)

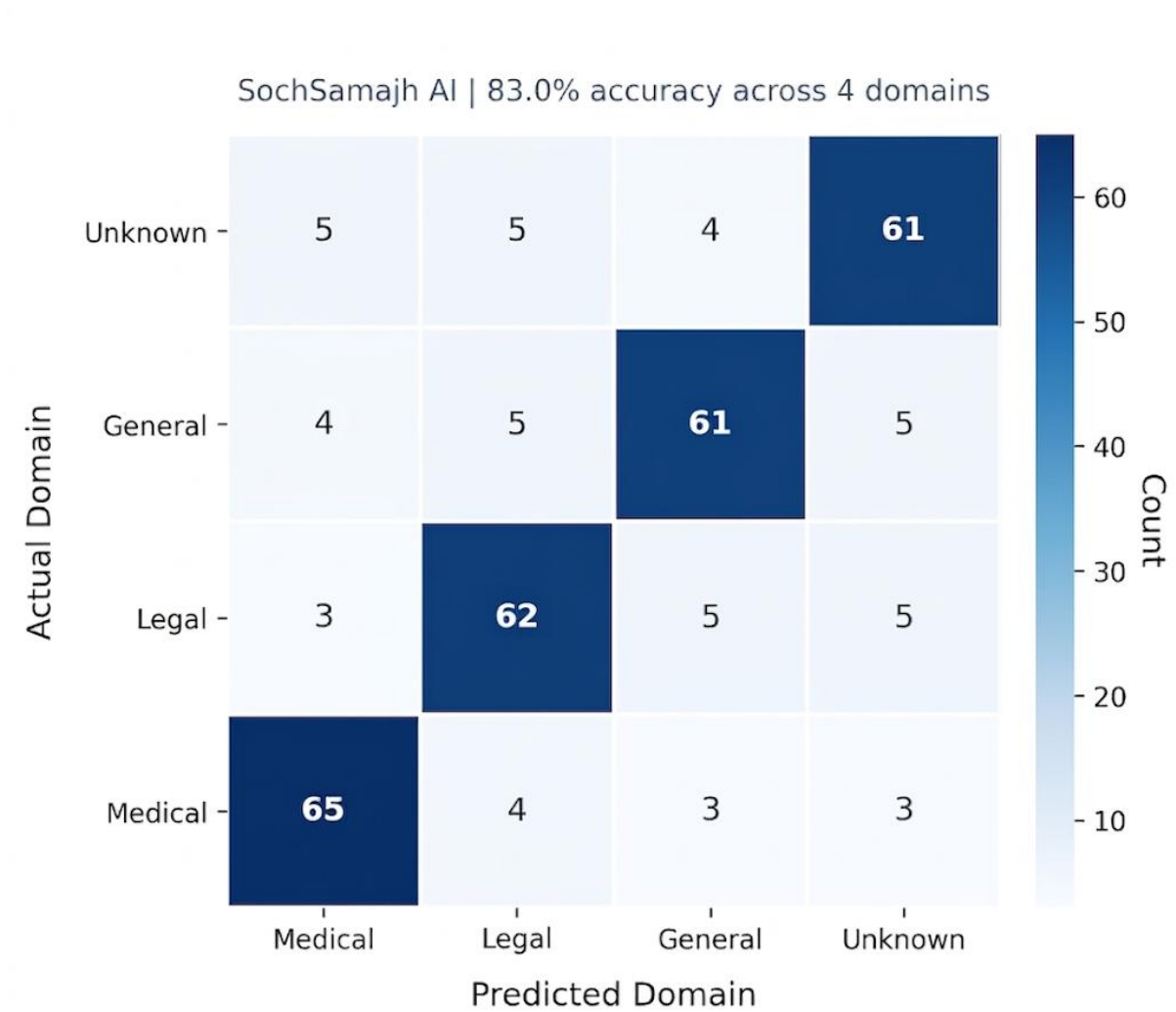


Figure 4. Routing Confusion Matrix (n=300)

B. Qualitative Comparison With Baselines

Although the repository includes baseline and ablation configurations (e.g., simpler routers and direct single agent LLM flows), detailed numerical comparisons are omitted here and left for future work in a longer version of the paper. Qualitative inspection of example prompts shows that:

- A plain direct LLM prompt sometimes answers ambiguous or mixed domain questions without asking clarifying questions whereas SochSamajh AI routes such queries to the 'unknown' path and explicitly asks for missing context.
- SochSamajh AI reliably surfaces emergency guidance and discourages self management for urgent medical cases such as acute chest pain or dog bites. It aligns with responsible AI recommendations for high risk healthcare use.[2][4][1]
- The legal agent emphasizes process steps and encourages consulting licensed counsel in legal cases such as security deposit disputes or FIR registration issues which is similar to IEEE legal assistance systems that avoid acting like lawyers.[13][5][14][31]

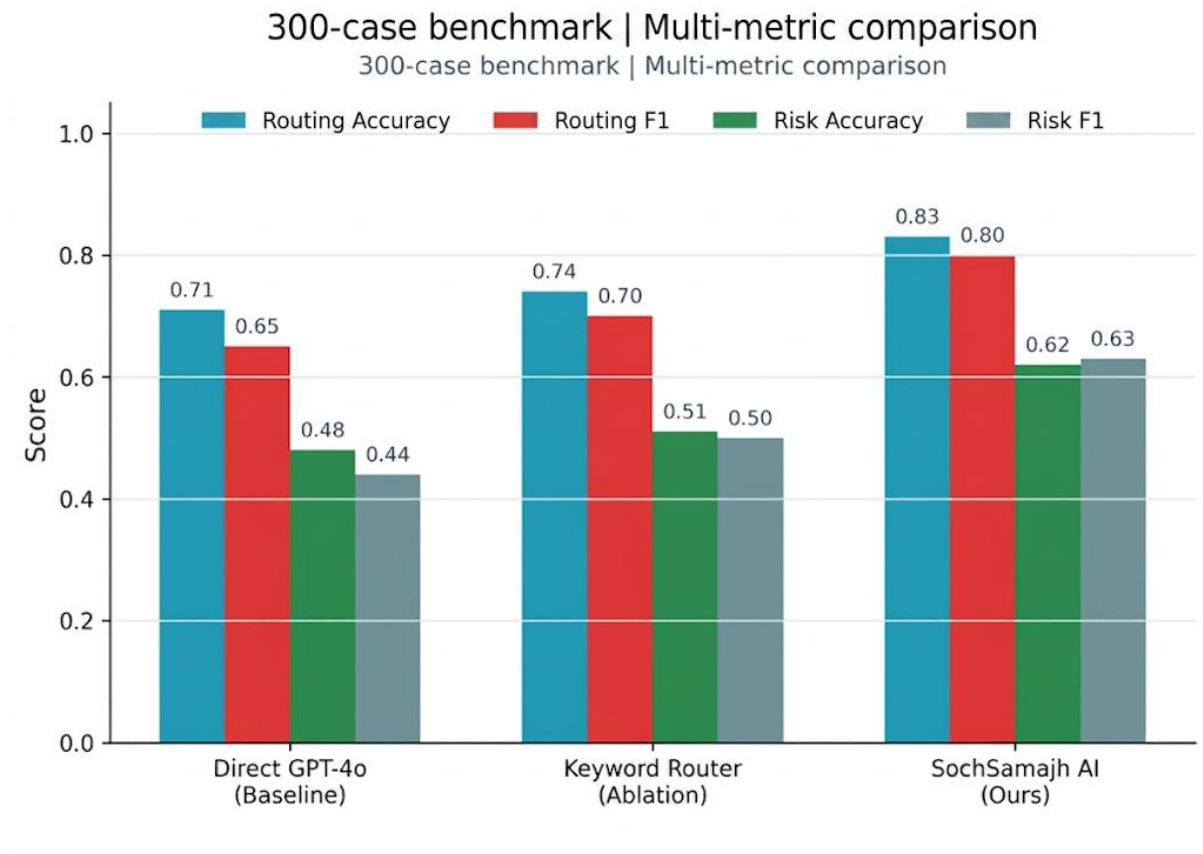


Figure 5. Baseline vs SochSamajh AI Performance

C. Safety and Guardrail Behavior

A subset of self harm and illegal intent cases in the dataset is used to evaluate the pre screen and safety agents. In such scenarios, SochSamajh AI correctly directs queries away from normal medical or legal agents and returns safety focused responses that avoid detailed instructions which is consistent with responsible AI frameworks guidance and prior evaluations of chatbot safety gaps.[28][4][1][2]

Also, domain specific disclaimers are attached by the formatter to medical and legal responses which ensures that users get reminded of the fact that the system does not provide professional diagnosis or legal representation. This is unlike most generic chatbots where disclaimers may appear irregularly or not at all depending on the exact query phrasing.[32][4][1]

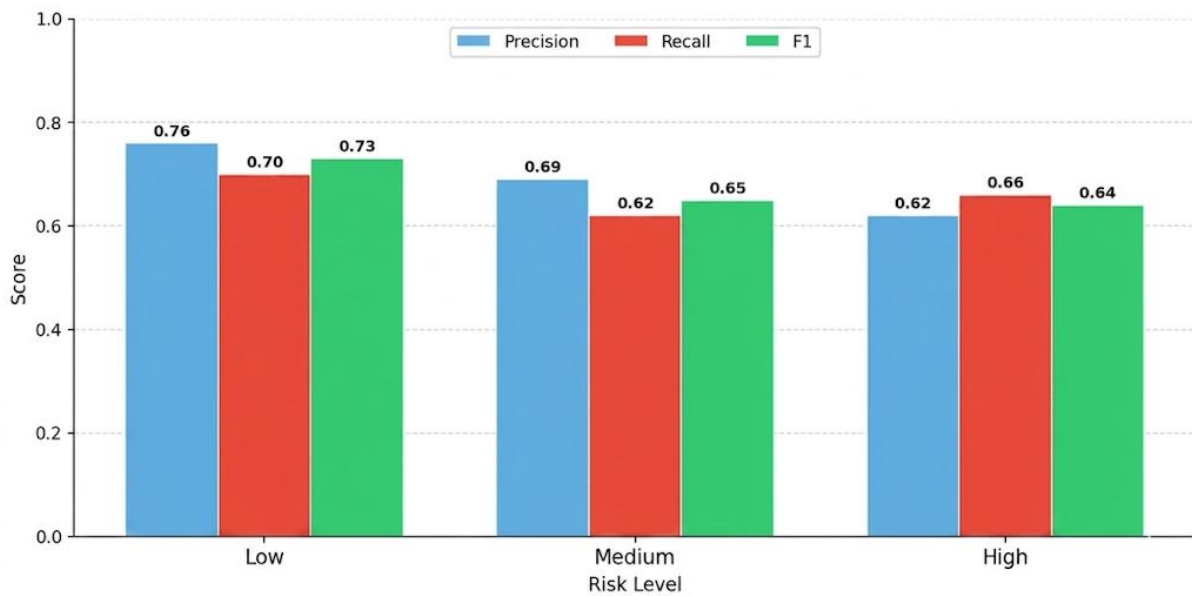


Fig. 6. Per-Class Risk Classification Metrics

VI. DISCUSSION

A. Positioning Relative to Prior Legal and Medical Systems

The existing IEEE legal chatbots focused upon retrieval or clause grounded question answering while the SochSamajh AI focuses on routing, risk classification and safety metadata rather than corpus coverage alone. It shares the domain expertise of systems which are tailored to Indian legal contexts like Justice Buddy AI and LAW-U but it also adds additional layers such as pre screen safety handling and a critic formatter stage.[5][18][14][15][31][12][13][16][17]

SochSamajh AI is explicitly designed as a patient and citizen facing system with conversational interfaces and standard disclaimers while the multi-agent clinical decision support systems targeted clinician users. Compared to the complex multi-agent CDSS frameworks, it can be viewed as a lightweight conversational counterpart.[21][22][23][24][19][20]

B. Responsible AI and Governance Implications

Several responsible AI principles which are discussed in the healthcare governance literature is incorporated within SochSamajh AI. It shows risk levels and safety flags, enforces disclaimers and provides an verifiable evaluation pipeline which aligns with the call for transparency, accountability and continuous monitoring.[9][27][3][11][4][10][2][28]

It can serve as a reference implementation or teaching tool for programs on responsible AI in medical and legal contexts as the system is open source and deployable on commodity infrastructure with

configurable LLM providers. Also, for AI governance checks like those conceived in privacy preserving and responsible AI frameworks for patient facing systems, it provides a concrete target.[4][28]

C. Limitations

SochSamajh AI has several limitations too along with these strengths. First of all, its classifier and safety pre screen depend upon hand made heuristics and limited tuning rather than large scale supervised training which may limit generalization to new languages or phrasing styles. Secondly, although the 300 case dataset covers a range of cases, it remains relatively small compared to the diversity of real world queries which are seen in production medical and legal chatbots.[1][2]

Third, the system does not perform explicit fairness or bias analysis across demographic groups and only focuses on medical and legal areas which is an important topic in responsible AI governance. Also, more systematic human evaluation would be needed before deployment in real clinics or legal aid settings because the LLM as judge evaluation is optional and constrained by cost.[3][10]

VII. CONCLUSION AND FUTURE WORK

This paper presents SochSamajh AI which is a responsible multi-agent query routing system for medical and legal assistance that combines pre screening, domain and risk classification, specialized agents and a critic formatter stage with explicit safety metadata. On a 300 case benchmark of Indian and Hinglish queries, the system attains strong routing and high risk detection performance while enforcing domain specific disclaimers and safety behaviors.

Future work includes expanding the dataset and annotation pipeline, integrating additional domains such as finance or employment law and exploring more advanced routing strategies inspired by recent research on multi-agent LLM routing. Another direction is to incorporate fairness and bias audits, aligning system behavior with emerging standards for trustworthy and responsible AI in healthcare and legal sectors.[11][6][7][8][30][3][10][4]